CONFAI 2026 · AI IN PRACTICE (EXTENDED TRACK) · POSTER EXHIBITION

Literature Discovery with Multi-Signal Ranking, Citation-Grounded Hybrid RAG, and 2D Novelty Geometry

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GitHub Code

Live Demo

① Multi-Signal Discovery

Ranks **5,752 catalog papers** and explores **101k citation works** via embeddings, 4-year half-life recency, citation saturation, and author prominence.

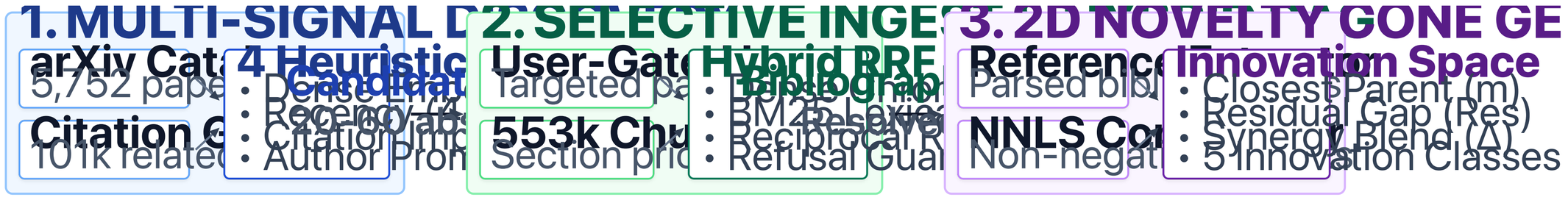
② Selective Ingest & Hybrid RAG

Scaling causes 41.9% MRR loss. Selective BM25 plus dense RRF recovers 6% to 61% of lost MRR with **98.89% citation validity**.

③ 2D Reference Cone Geometry

NNLS cone projection decomposes innovation into Closest Parent (m), Synergy Blend (Δ), and Residual Gap (Res).

System Architecture: Literature Discovery, Selective Grounded RAG, and Innovation Subspaces

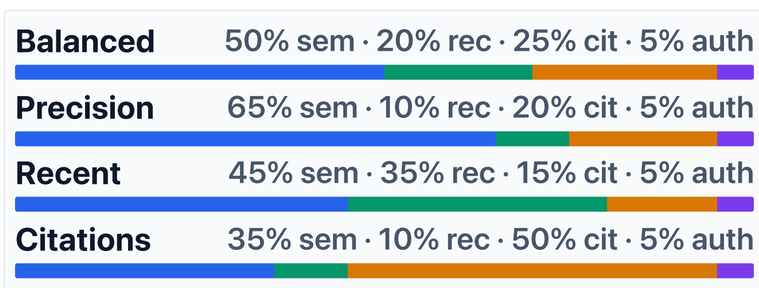


1. Multi-Signal Discovery

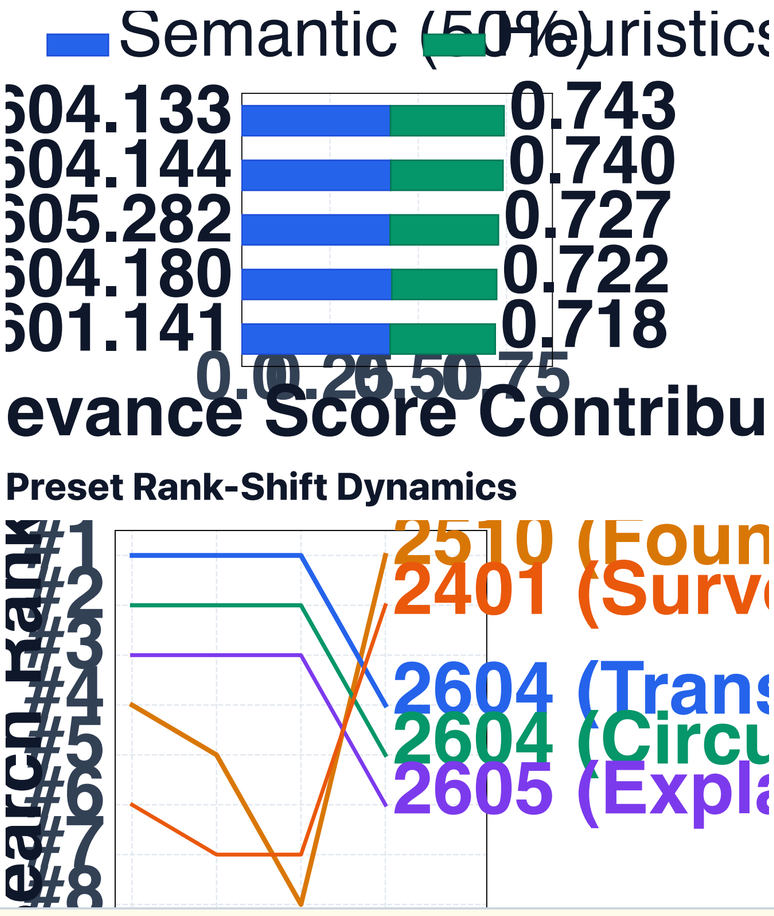
HEURISTIC RANKING & PRESET SHIFTS

```
Score(q, d) = w_sem · S_sem + w_rec · 0.5^(Δt/4)
  + w_cit · C_score + w_auth · A_score
```

Weight Presets Breakdown



Top-5 Worked Query: "vision transformer"

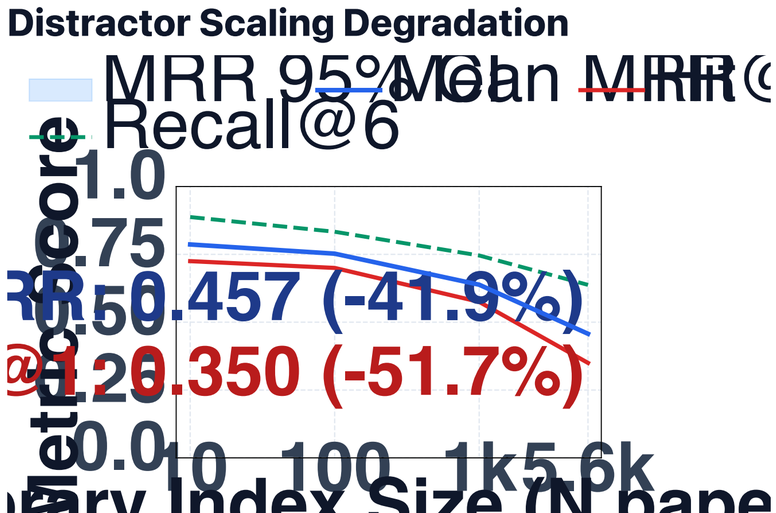


Preset Rank-Shift Dynamics

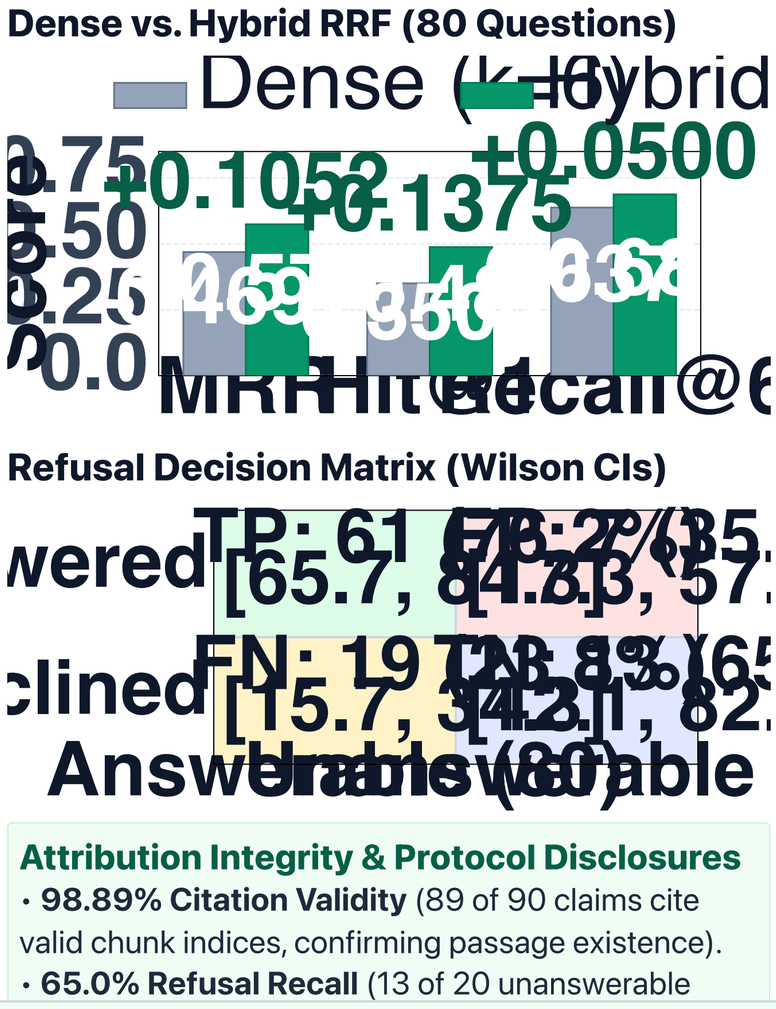
2. Hybrid RRF RAG

SCALING DEGRADATION & ATTRIBUTION

Distractor Scaling Degradation



Dense vs. Hybrid RRF (80 Questions)



Refusal Decision Matrix (Wilson CIs)

Attribution Integrity & Protocol Disclosures

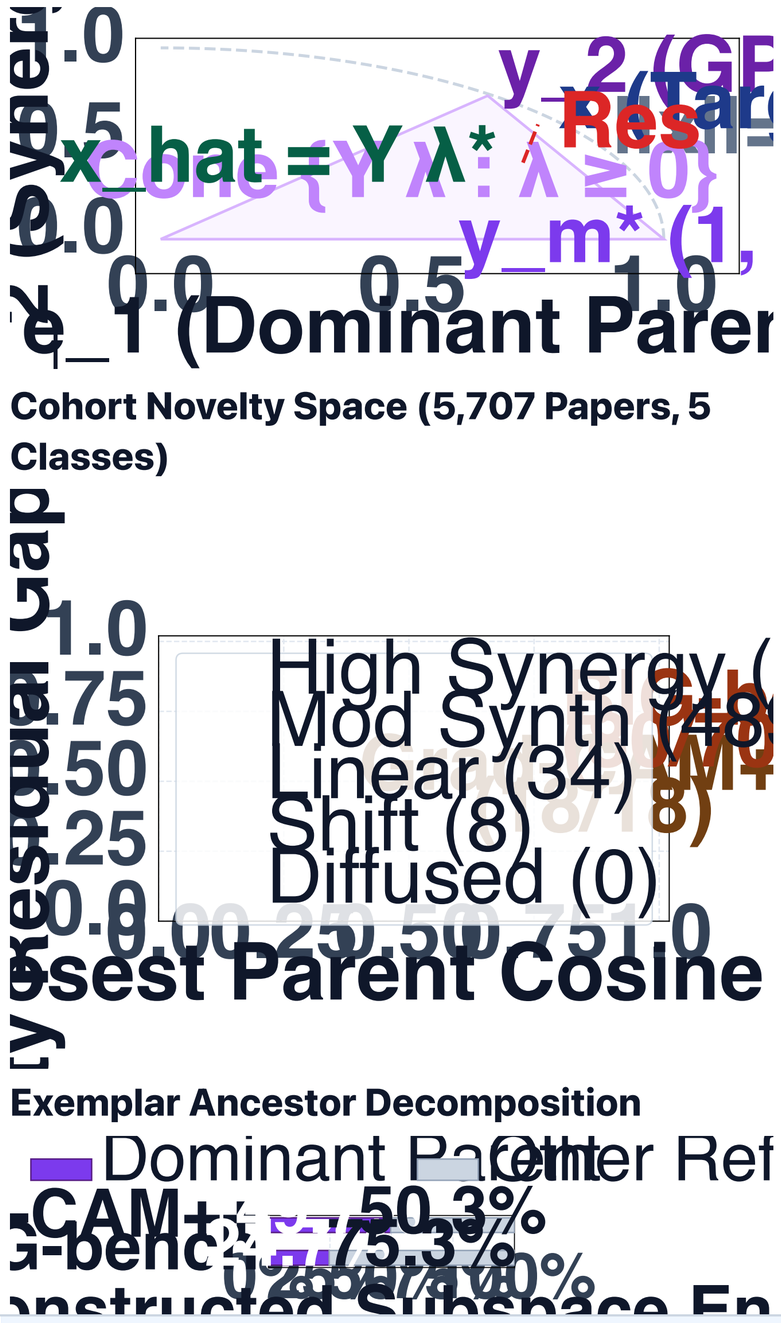
- **98.89% Citation Validity** (89 of 90 claims cite valid chunk indices, confirming passage existence).
- **65.0% Refusal Recall** (13 of 20 unanswerable

3. 2D Novelty Geometry

SUBSPACE INNOVATION & NULL CONTROL

```
λ* = argmin(λ ≥ 0) || x − Y λ ||²,
    x_hat = Y λ*
m = max_j (x^T y_j),    Δ = R²_+ − max(m, 0)²,    Res = √(1 − R²_+)
```

2D Reference Cone Geometry Canvas



Cohort Novelty Space (5,707 Papers, 5 Classes)

Exemplar Ancestor Decomposition

Limitations (AI in Practice)

- **Refusal:** Declines 23.8% of queries under weak evidence.
- **Lexical:** Dense search misses exact theorem IDs without BM25.
- **MPS Memory:** 1 GB embedding weights require proactive cache release.

Conclusions

- **Gating:** Scaling drops Hit@1 by 51.7%, confirming gating is vital.
- **Hybrid:** Reciprocal rank fusion yields paired MRR gain of +0.1052.
- **Synergy:** Real paper synergy exceeds isotropic null baseline by 5.0x.

Future Work & Contact

- **Learned Priors:** Query-dependent section attention weights.
- **Lineage:** Multi-hop semantic traversal and co-citations.
- **Contact:** Ajay Sharma, Dr. Subrat Panda (Plaksha University).

References: [1] Vaswani et al., Attention, NeurIPS 2017. [2] Cormack et al., RRF, SIGIR 2009. [3] Selvaraju et al., Grad-CAM, ICCV 2017. [4] Srivastava et al., BIG-bench, 2022.

* Snapshots: N=5,657 (final_eval.json) vs N=5,663 (sweep). RRF (k=60, α=0.5) and section priors were heuristic defaults, not tuned on eval questions. Gold questions LLM-generated against 10-paper index.